CS3 RESEARCH EXPERIENCE FOR TEACHERS WEEK 2

Concepts in Deep Learning

Iterations, epochs, batch size, and overfitting

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OVERVIEW

Four words you will hear in every training run

Iteration

One update of the model's parameters.

Batch size

The number of training examples used in one iteration.

Epoch

One complete pass through the entire training dataset.

Overfitting

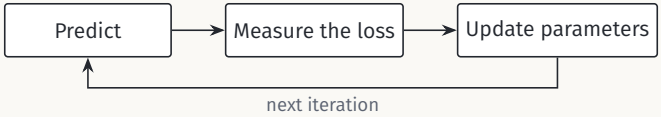
Learning the training data so closely that the model fails on new data.

An iteration is one update of the parameters

- The model makes predictions, measures its error with a loss, and adjusts its parameters a little.
- Training repeats this step thousands of times.
- Each step should leave the model slightly better than before.

Everyday analogy: cooking a new recipe

Each time you cook the dish, you make a small adjustment: less salt, a longer bake. You are updating the parameters of your technique, and the dish improves over many attempts.



An epoch is one full pass through the dataset

- After one epoch, the model has seen every training example once.
- Training usually runs for many epochs.
- Each pass reinforces what the model learned in the previous one.

Everyday analogy: studying from a textbook

Reading the whole textbook once is one epoch. You might read it several times before the exam, and each complete read-through helps you learn the material better.

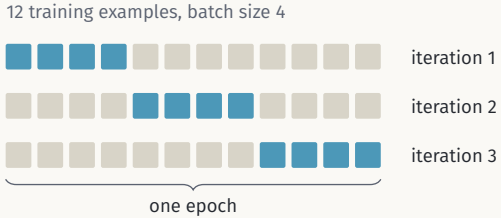
Batch size sets how many examples share one update

- Small batches give fast, frequent, but noisier updates.
- Large batches give smoother updates but need more memory and time per step.
- The batch size also fixes how many iterations make up one epoch.

Everyday analogy: doing laundry

Your laundry is the dataset, and the batch size is how many clothes go into one load. Small loads finish quickly but take many trips; large loads take longer each time.

Putting the three together



The relationship

$$\text{iterations per epoch} = \left\lceil \frac{\text{number of training examples}}{\text{batch size}} \right\rceil$$

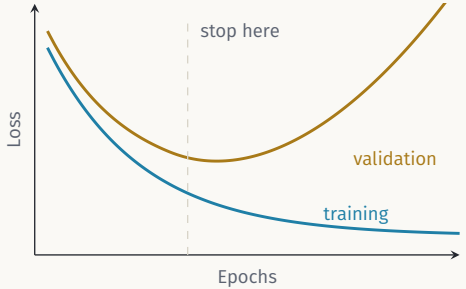
With 50,000 images and a batch size of 100, one epoch takes 500 iterations; 20 epochs take 10,000.

Overfitting: memorizing instead of learning

- The model learns the training data too well, including its noise.
- Training loss keeps falling while validation loss starts to rise.
- We stop training, or add data and regularization, at the point where validation loss is lowest.

Everyday analogy: memorizing for a test

You memorize specific questions and answers and do well on those exact questions, but struggle with new, slightly different ones.



Illustrative curves.

Summary

- **Iteration**: one parameter update, computed from one batch.
- **Batch size**: how many examples are used in that update.
- **Epoch**: one full pass through the training set, made of many iterations.
- **Overfitting**: strong training performance that does not carry over to new data; watch the validation loss to catch it.